

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic and modern visual effect.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

ARC209-R

Cost-optimization strategies for your non-production workloads

John Lee

(he/him)

Sr. Solutions Architect

AWS

Eugene Wiehahn

(he/him)

Sr. Solutions Architect

AWS



© 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda

- Identifying non-production workload spend
- Establishing ongoing cost governance
- Scenarios
- Non-production cost optimization strategies

What is non-production?

- Development
- Test
- Staging
- QA
- Pre-production
- Any workload that does not serve live customer traffic

**Do you know how much your company
saved by migrating to AWS?**



What the data says

Accenture found that moving workloads to the public cloud led to total cost of ownership (TCO) savings of 30–40%¹

**Are you able to distinguish
non-production spend from
production spend?**

What the data says

Only 3 out of 10 organizations
know exactly where their cloud
costs are going¹

Getting there can be challenging



**Tagging is like exercise.
Everyone agrees it should be done . . .
but it is often hard to implement and
keep consistent.**

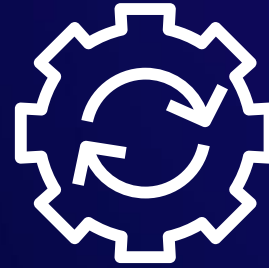
Awareness: Resource tagging



Technical tags



Business tags



Tags for
automation



Security tags

Tagging = accountability

Resource tagging: Awareness

Preventative

Block the deployment of non-conforming resources

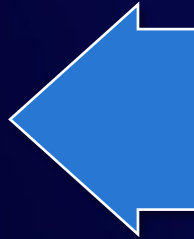


AWS tools

Service control and tag policies, IAM policies

Proactive

Catch non-conforming resources during the development

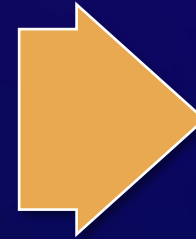


AWS tools

AWS CloudFormation Guard and hooks, AWS Service Catalog

Reactive

Take action on existing non-compliant resources



AWS tools

Tag Editor, AWS Config, AWS Security Hub, AWS Lambda

**What percentage of your AWS spend
is on your non-production workloads?**

What the data says

44% of public cloud-based workloads are classified as non-production workloads¹

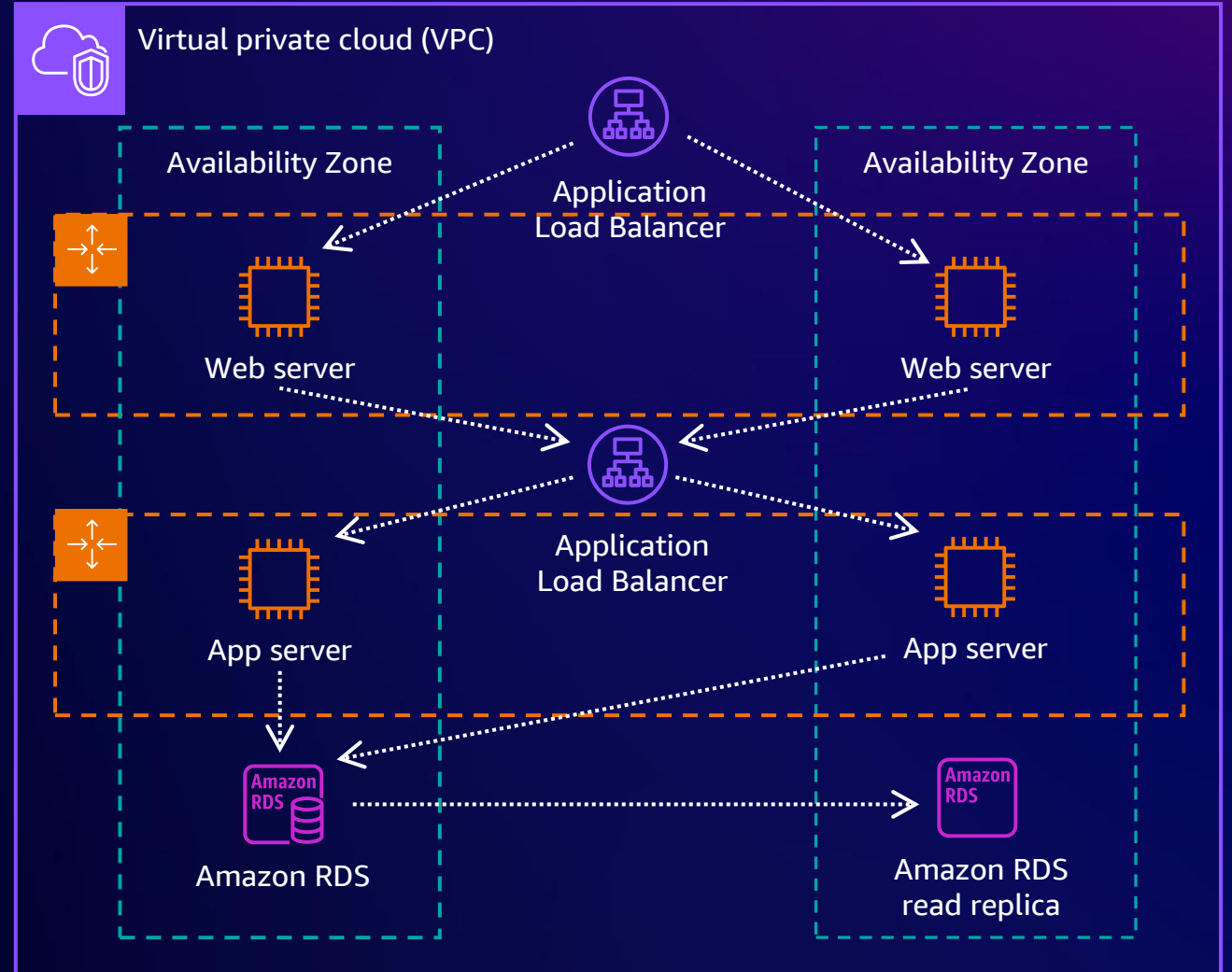
Scenarios



Scenario 1a

You are testing your application code changes for your web application architecture in your development environment.

What would you do differently from your production to minimize spend?



Scenario 1b

After you have run your tests, you deleted the Amazon EC2 instances, Application Load Balancers, and the Amazon RDS database. However, you are still incurring costs. What could be the cause?

Scenario 2

You have large amounts of production data (500+ TB) stored in your Amazon Aurora database, Amazon S3 bucket, and Amazon Redshift cluster. You are required to run tests on the same set of data. How would you do it while minimizing costs?

Scenario 3

You have hundreds of developers who are all demanding that they have their own AWS accounts for testing. You would like to make sure that your developers does not spend over \$500 per month and want to prevent developers from using GPU instances. How would you accomplish this?

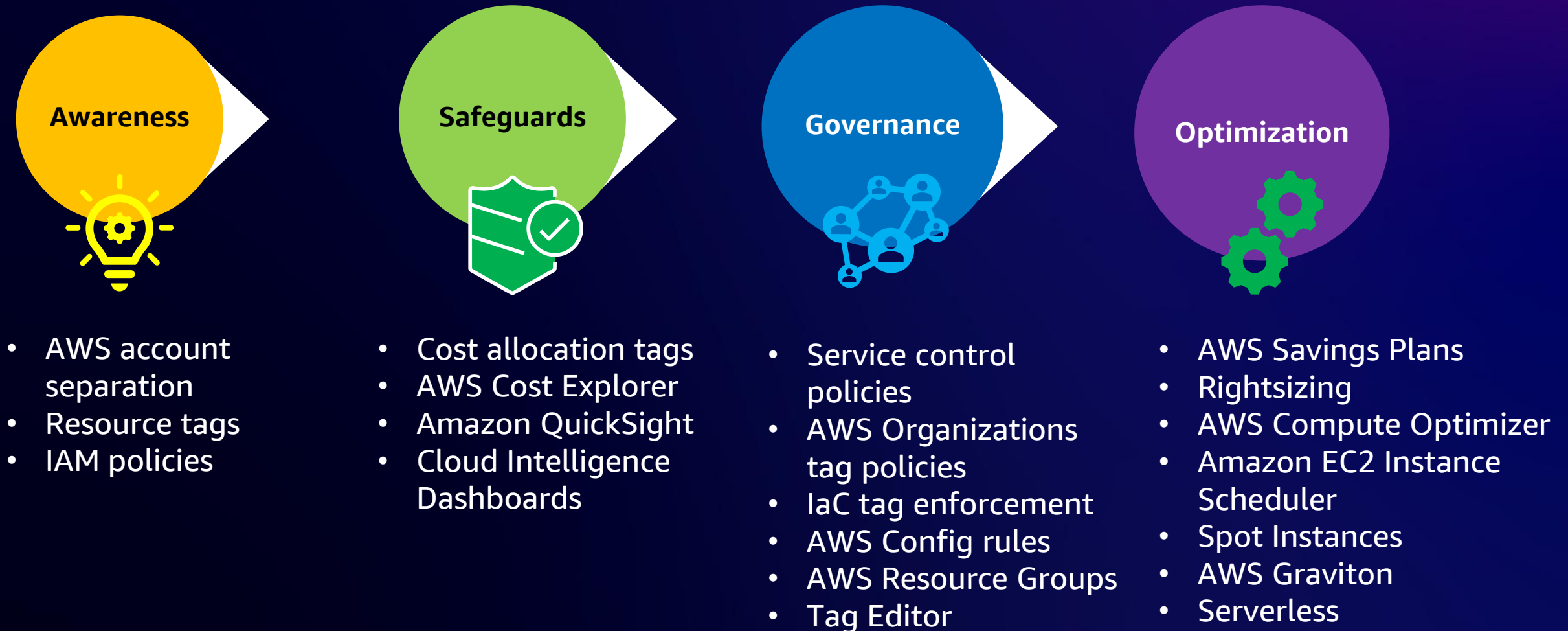
Scenario 4

You have a single AWS account that hosts both production and non-production workloads. You would like to move your non-production workloads to a separate AWS account. Where would you start?

Scenario 5

You would like to delete most resources in your test AWS account except a few select resources every night. How would you do this?

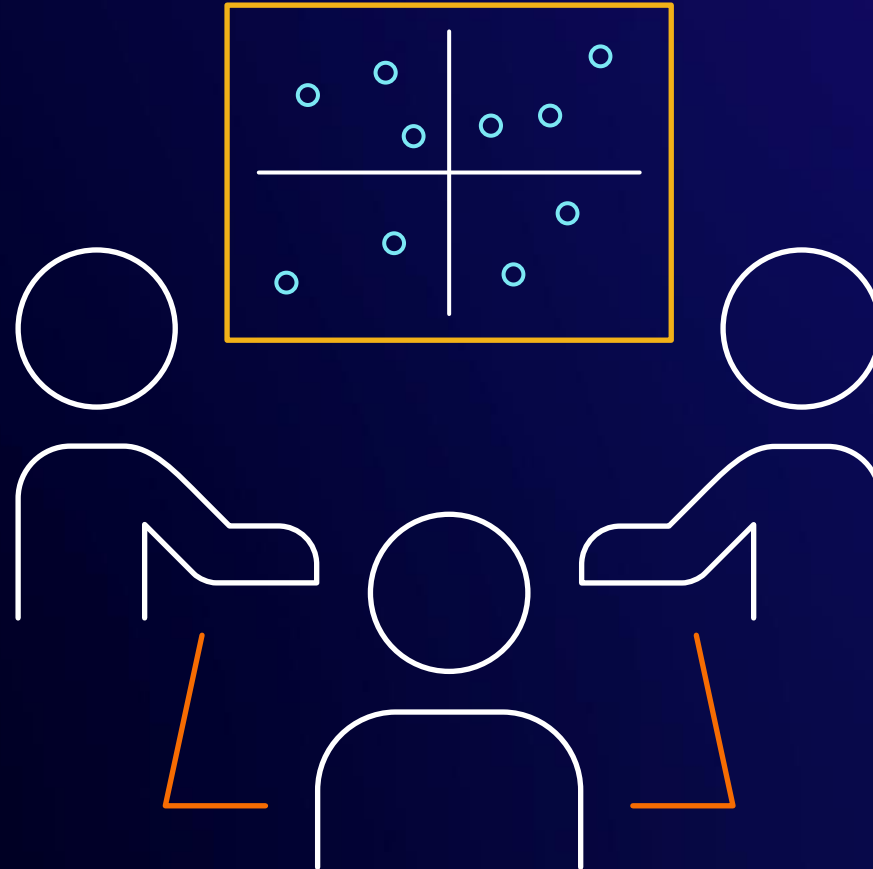
The final state



Working with you to optimize savings in non-production environments

AWS Cloud Financial Management services

Cost transparency, control, forecasting, and optimization help



AWS Well-Architected

Hands-on labs, videos, and workshops ensure the most cost-effective resources every time

Thank you!

Eugene Wiehahn
wiehahne@amazon.com

John Lee
leemjohn@amazon.com



Please complete the session
survey in the mobile app

